

Supplemental Material

for

Methodological Flexibility in the Iowa Gambling Task Undermines Interpretability: A Meta-method Review

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Table S1.*Iowa Gambling Task (IGT) Meta-Analyses Examined for Procedural and Scoring Method Criteria*

ID	Short Reference	Cited by (2024)	Cited by (2026)
M1	Richard-Devantoy et al., 2014	222	251
M2	Kovács et al., 2017	90	121
M3	Stephan et al., 2017	76	100
M4	Guillaume et al., 2015	73	92
M5	Biernacki et al., 2016	61	76
M6	Richard-Devantoy et al., 2016	42	49
M7	Paret et al., 2017	34	46
M8	Kluwe-Schiavon et al., 2020	34	48
M9	Evens et al., 2016	32	42
M10	Perrain et al., 2021	29	49
M11	Mukherjee & Kable, 2014	28	33
M12	Rotge et al., 2017	26	37
M13	Edge et al., 2013	25	30
M14	Woodrow et al., 2019	23	32
M15	Sastre-Buades et al., 2021	23	42

Note. This table lists the 15 meta-analyses on the IGT that were screened to assess their reported inclusion and exclusion criteria for procedural and scoring methods. One additional meta-analysis (Howard et al., 2020) was excluded as it did not include any primary studies using the IGT. The 'ID' column provides a unique identifier for each meta-analysis used for reference within this article. Citation counts were retrieved from Scopus. The 'Cited by (2024)' and 'Cited by (2026)' numbers reflect the counts as of April 23, 2024, and January 08, 2026.

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- M6: Richard-Devantoy, S., Olié, E., Guillaume, S., & Courtet, P. (2016). Decision-making in unipolar or bipolar suicide attempters. *Journal of Affective Disorders*, 190, 128–136.
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Table S2.*Interrater Reliability for Procedural Variables*

Variable	Cohen's Kappa (κ)	SE (Bootstrap)	95% CI (low)	95% CI (high)
sample size	0.76	0.04	0.68	0.84
variant	0.88	0.06	0.76	1.00
protocol	0.55	0.08	0.40	0.71
goal	0.69	0.08	0.53	0.82
instructions	0.44	0.09	0.26	0.61
feedback	0.57	0.08	0.41	0.71
incentives	0.53	0.07	0.40	0.66
currency	0.75	0.05	0.65	0.86
starting Capital	0.80	0.06	0.69	0.91
reward per card (advantageous decks)	0.65	0.06	0.52	0.77
reward per card (disadvantageous decks)	0.63	0.06	0.51	0.75
net gain advantageous deck 1	0.70	0.06	0.57	0.80
net gain advantageous deck 2	0.71	0.06	0.58	0.82
net loss disadvantageous deck 1	0.69	0.06	0.57	0.81
net loss disadvantageous deck 2	0.68	0.06	0.55	0.79
loss interval advantageous deck 1	0.73	0.06	0.60	0.85
loss interval advantageous deck 2	0.73	0.07	0.60	0.86

Variable	Cohen's Kappa (κ)	SE (Bootstrap)	95% CI (low)	95% CI (high)
loss interval disadvantageous deck 1	0.73	0.07	0.60	0.87
loss interval disadvantageous deck 2	0.73	0.07	0.59	0.85
loss frequency advantageous deck 1	0.70	0.07	0.56	0.82
loss frequency advantageous deck 2	0.72	0.07	0.59	0.85
loss frequency disadvantageous deck 1	0.69	0.07	0.56	0.82
loss frequency disadvantageous deck 2	0.72	0.06	0.60	0.84
number of blocks	0.80	0.05	0.69	0.90
number of trials	0.90	0.06	0.76	1.00

Note. SE = standard error; CI = confidence interval. The standard error was calculated using a bootstrapping method with N = 1,000 iterations.

Table S3.*Underreporting, as Indicated by Indeterminable Values per Variable*

Variable	Clinical Studies (N)	Clinical Studies (%)	Non- clinical Studies (N)	Non- clinical Studies (%)
task variant	45	84.91	37	69.81
number of trials	6	11.32	2	3.77
number of trial blocks	2	3.77	0	0.00
instructions	45	84.91	32	60.38
goal	24	45.28	13	24.53
feedback	43	81.13	31	58.49
currency	27	50.94	12	22.64
amount of starting capital	39	73.58	23	43.40
reward per card (advantageous decks)	38	71.70	27	50.94
reward per card (disadvantageous decks)	37	69.81	27	50.94
net gain advantageous deck 1	36	67.92	29	54.72
net gain advantageous deck 2	36	67.92	28	52.83
net loss disadvantageous deck 1	36	67.92	29	54.72
net loss disadvantageous deck 2	36	67.92	28	52.83
loss frequency advantageous deck 1	41	77.36	28	52.83
loss frequency advantageous deck 2	40	75.47	27	50.94
loss frequency disadvantageous deck 1	41	77.36	27	50.94
loss frequency disadvantageous deck 2	40	75.47	27	50.94
interval for infrequent losses	38	71.70	27	50.94
incentives	36	67.92	14	26.42
TOTAL	800	65.63	549	45.04

Note. For each variable, the table shows the number (N) and percentage (%) of IGTs in clinical (N = 53) and non-clinical (N = 53) studies in which information was reported in a way that was indeterminable. The 'TOTAL' row summarizes the sum of all indeterminable instances for each group. The row 'interval for infrequent losses' summarizes four variables (one per deck) with identical counts; the 'TOTAL' row counts all four, so it exceeds the sum of the rows shown. IGT = Iowa Gambling Task.

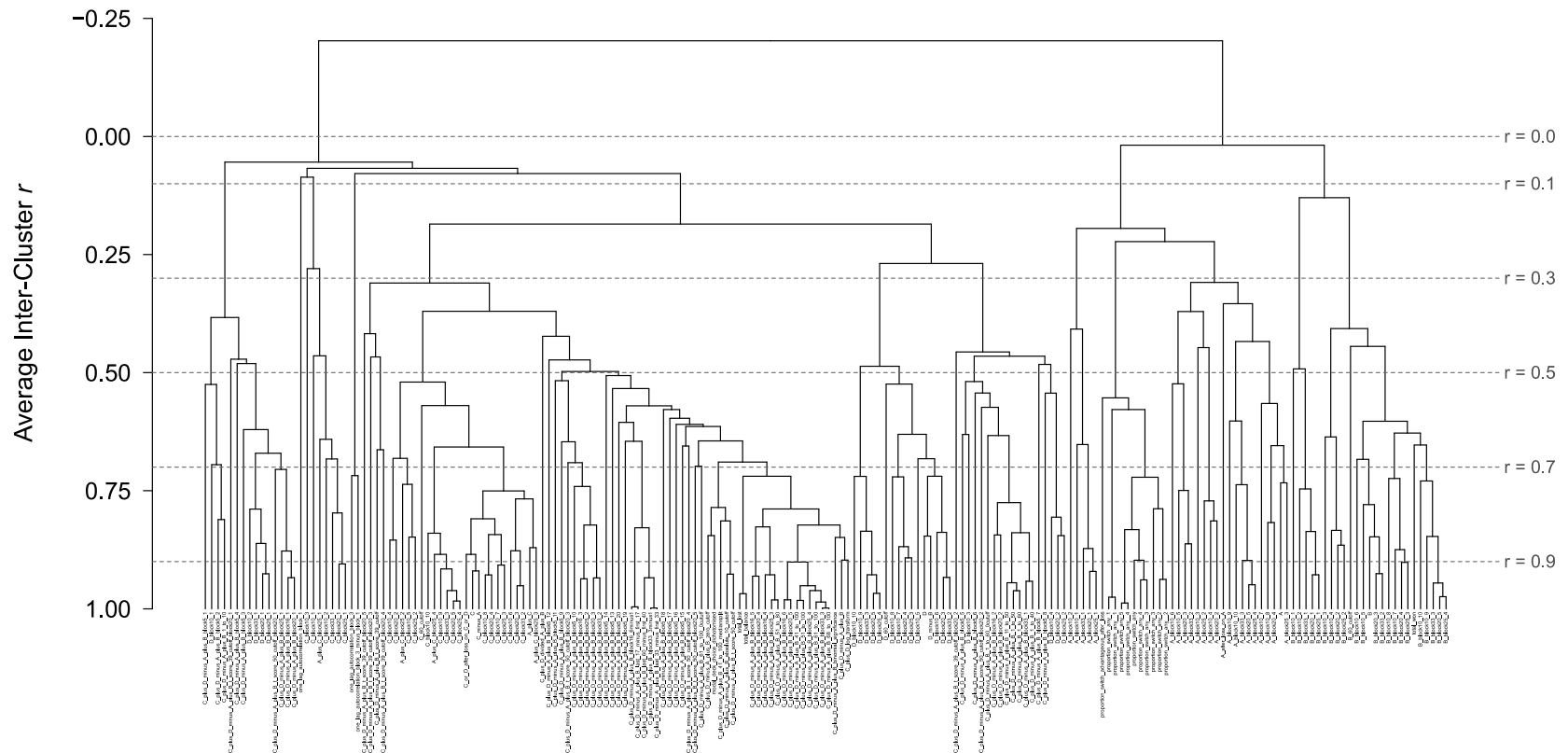
Table S4.*Cluster Solutions at Fixed Correlation Thresholds and Under the Dynamic Hybrid Cut*

Cut method	<i>k</i> clusters	Min within-cluster mean <i>r</i>	Max between-cluster mean <i>r</i>
Threshold <i>r</i> = 0.9	152	0.902	0.898
Threshold <i>r</i> = 0.7	85	0.718	0.698
Threshold <i>r</i> = 0.5	37	0.629	0.497
Threshold <i>r</i> = 0.3	13	0.403	0.280
Threshold <i>r</i> = 0.1	7	0.317	0.086
Threshold <i>r</i> = 0.0	2	0.187	-0.202
Dynamic hybrid cut	20	0.418	0.641

Note. The dynamic hybrid cut (deepSplit = 2, minClusterSize = 5) yielded 20 clusters with zero unassigned variables. At *r* = 0.9 the great majority of the 196 strategies remain unique, indicating few near-exact duplicates. The space consolidates slowly and splits into two directionally opposed mega-clusters at *r* = 0.0.

Figure S1.

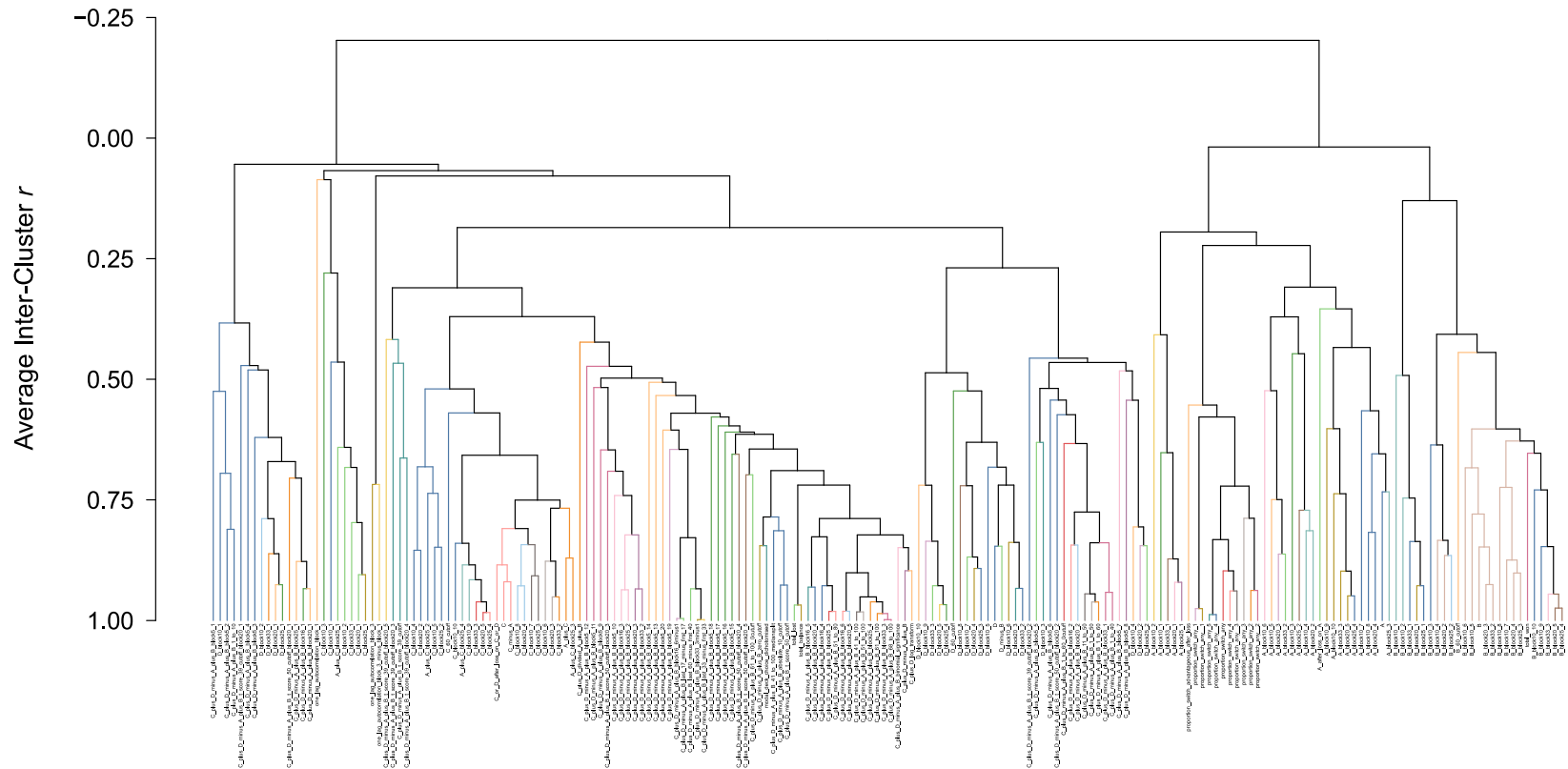
Average-linkage hierarchical clustering of IGT scoring methods based on pairwise Pearson correlations.



Note. Dendrogram of 196 IGT scoring methods derived from 504 participants, clustered using average linkage on distance = $1 - r$. The y-axis shows the average Pearson correlation between the two groups joined at each merge step; higher branches reflect less similar groupings. Dashed horizontal lines mark reference thresholds at $r = 0.0, 0.1, 0.3, 0.5, 0.7,$ and 0.9 .

Figure S2.

Dynamic Hybrid Cut cluster assignments overlaid on the average-linkage dendrogram.



Note. The dendrogram structure is identical to Figure S1. Branch colours indicate cluster membership as identified by the Dynamic Hybrid Cut algorithm (deepSplit = 2, minClusterSize = 5). 20 clusters were identified, and all 196 scoring methods were assigned to a cluster; black branches join different clusters.